Cost-Optimization Model for Pipe Weight Reduction

Saint-Gobain Pipeline (Maanshan) Co., ltd
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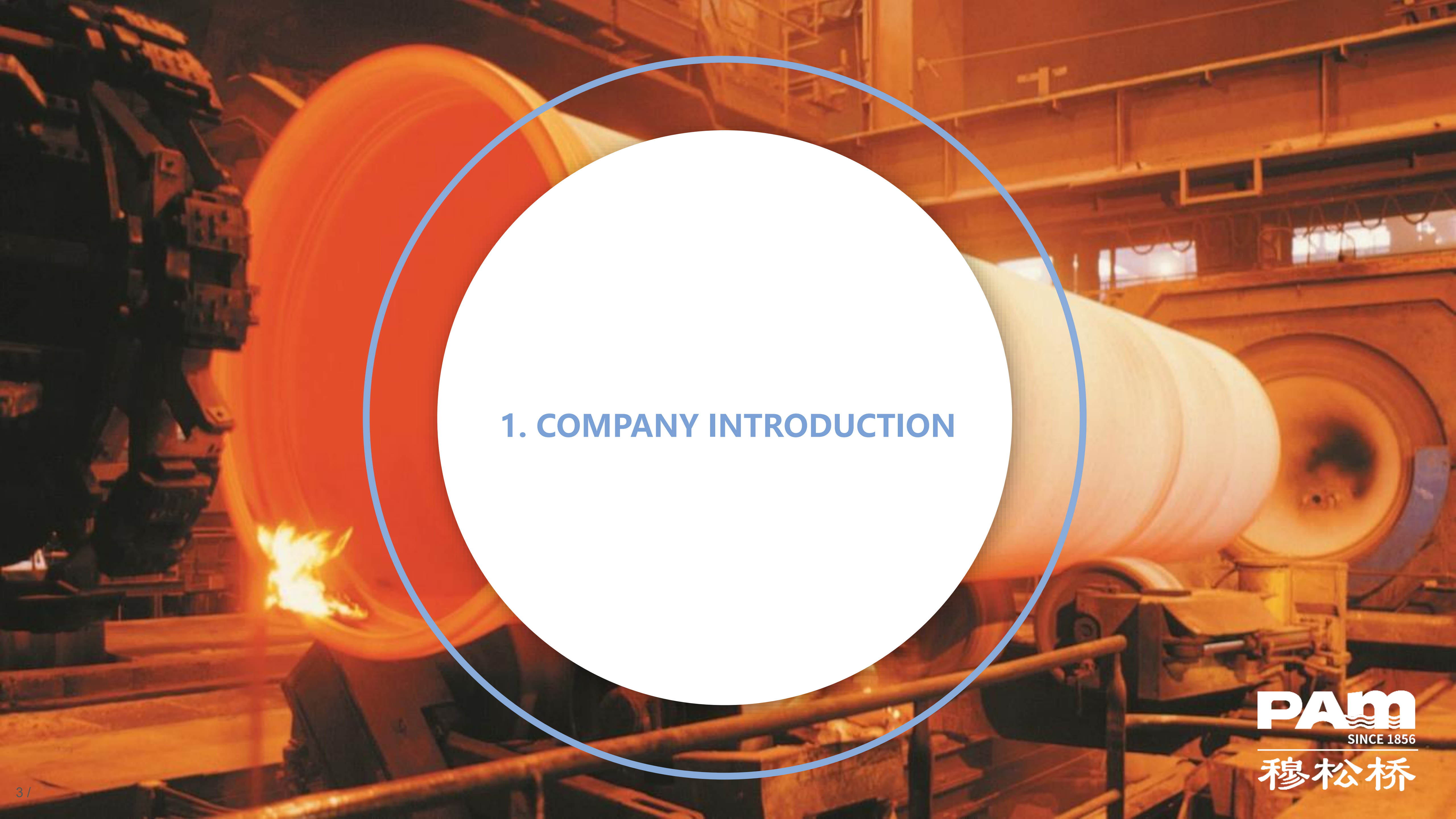
PAM SINCE 1856 穆松桥



1. COMPANY INTRODUCTION

SAINT-GOBAIN GROUP: THE LEADING IN BUILDING CONSTRUCTION FIELD

French company; Headquarter: Paris; Founded in 1665年，2021 the world 500 fortune 266^{th}

History **357** years

Business **68** countries

World wide >**71,000** employees

From >**100** countries

4,000

sales agencies

2021sales revnue

38 billion Euros

>**75%** sales revenue comes from sustainable construction market

World wide industry company

top **100** , >1000 manufacturing sites

Worldwide construction material

Top **100**

SAINT-GOBAIN

PAM SINCE 1856 穆松桥

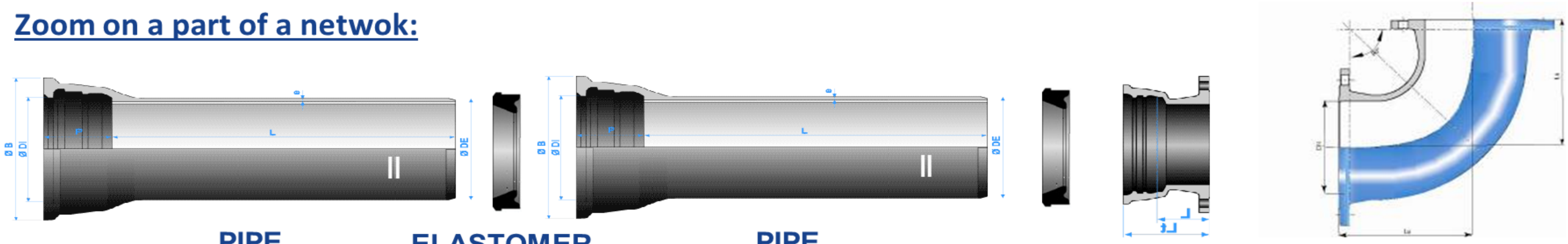
PAM: FOUNDED IN 1856, A BRANCH OF SG GROUP

Pont-a-Mousson=PAM; Headquarter: PAM; Founded in 1856，leading in ductile iron pipe

Product introduction

Zoom on a part of a netwok:

ELASTOMER FLANGE GASKET



PIPE

ELASTOMER GASKET

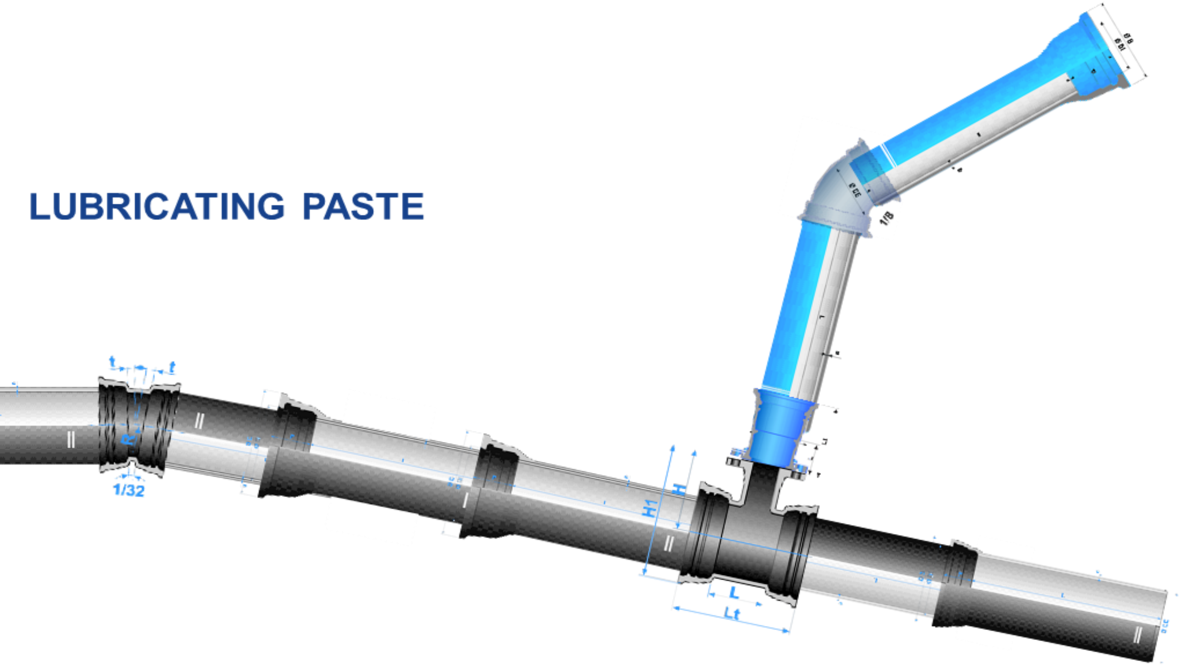
PIPE

FLANGE SOCKET

BEND ¼ FF

LUBRICATING PASTE

A PIPELINE



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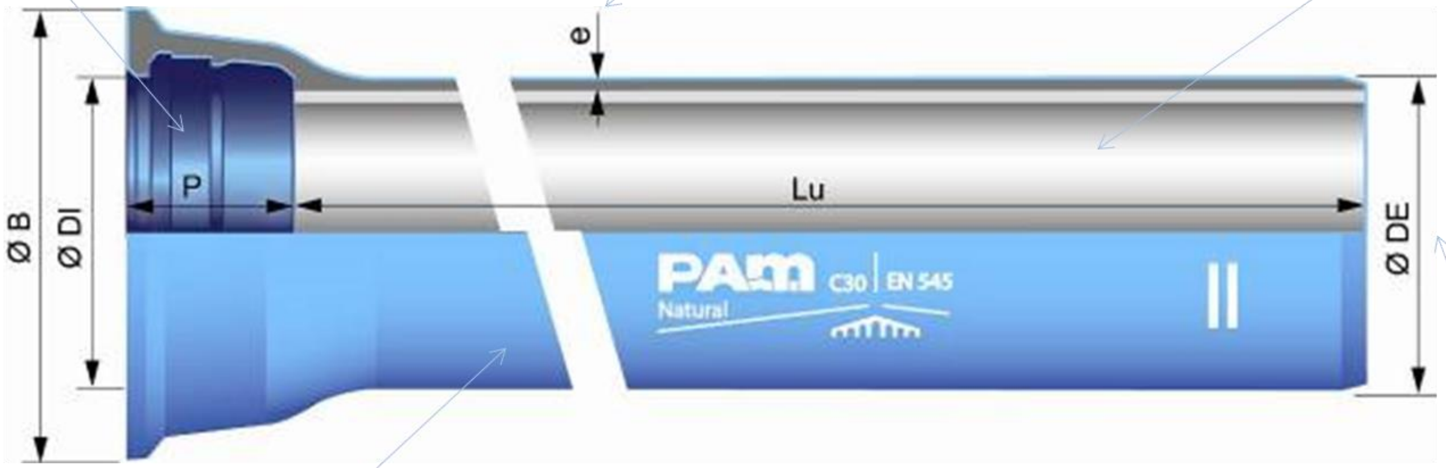
Product introduction

Zoom on a pipe:

Thickness

Type of joint

Internal lining



External coating

Nominal diameter

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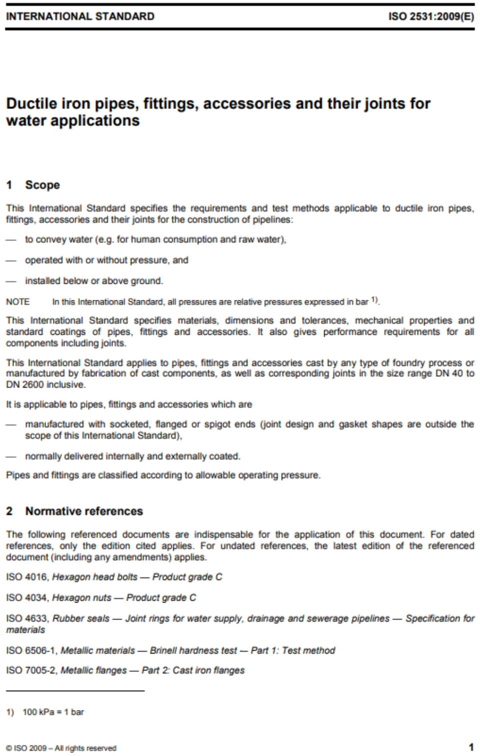
Product Standard

- GB/T13295; ISO 2531; EN545
  Ductile iron pipes, fittings and accessories for water or gas applications

- GB/T 26081; ISO 7186; EN877
  Ductile iron pipes, fittings and accessories for sewerage applications



INTERNATIONAL STANDARD — ISO 2531:2009(E)

Ductile iron pipes, fittings, accessories and their joints for water applications

1 Scope

This International Standard specifies the requirements and test methods applicable to ductile iron pipes, fittings, accessories and their joints for the construction of pipelines:

— to convey water (e.g. for human consumption and raw water),

— operated with or without pressure, and

— installed below or above ground.

NOTE In this International Standard, all pressures are relative pressures expressed in bar 1).

This International Standard specifies materials, dimensions and tolerances, mechanical properties and standard coatings of pipes, fittings and accessories. It also gives performance requirements for all components including joints.

This International Standard applies to pipes, fittings and accessories cast by any type of foundry process or manufactured by fabrication of cast components, as well as corresponding joints in the size range DN 40 to DN 2600 inclusive.

It is applicable to pipes, fittings and accessories which are

— manufactured with socketed, flanged or spigot ends (joint design and gasket shapes are outside the scope of this International Standard),

— normally delivered internally and externally coated.

Pipes and fittings are classified according to allowable operating pressure.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 4016, *Hexagon head bolts — Product grade C*

ISO 4034, *Hexagon nuts — Product grade C*

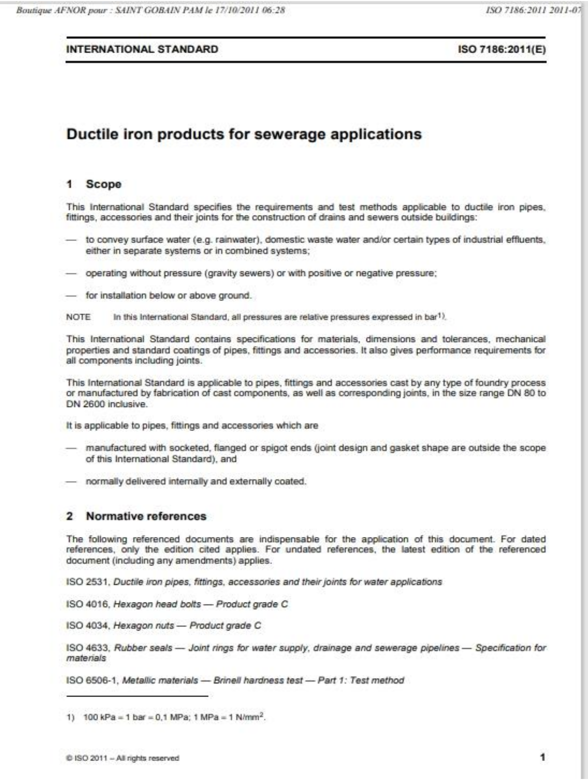
ISO 4633, *Rubber seals — Joint rings for water supply, drainage and sewerage pipelines — Specification for materials*

ISO 6506-1, *Metallic materials — Brinell hardness test — Part 1: Test method*

ISO 7005-2, *Metallic flanges — Part 2: Cast iron flanges*

1) 100 kPa = 1 bar

 1



Boutique AFNOR pour : SAINT GOBAIN PAM le 17/10/2011 06:28 *ISO 7186:2011 2011-07*

INTERNATIONAL STANDARD — ISO 7186:2011(E)

Ductile iron products for sewerage applications

1 Scope

This International Standard specifies the requirements and test methods applicable to ductile iron pipes, fittings, accessories and their joints for the construction of drains and sewers outside buildings:

— to convey surface water (e.g. rainwater), domestic waste water and/or certain types of industrial effluents, either in separate systems or in combined systems;

— operating without pressure (gravity sewers) or with positive or negative pressure;

— for installation below or above ground.

NOTE In this International Standard, all pressures are relative pressures expressed in bar 1).

This International Standard contains specifications for materials, dimensions and tolerances, mechanical properties and standard coatings of pipes, fittings and accessories. It also gives performance requirements for all components including joints.

This International Standard is applicable to pipes, fittings and accessories cast by any type of foundry process or manufactured by fabrication of cast components, as well as corresponding joints, in the size range DN 80 to DN 2600 inclusive.

It is applicable to pipes, fittings and accessories which are

— manufactured with socketed, flanged or spigot ends (joint design and gasket shape are outside the scope of this International Standard), and

— normally delivered internally and externally coated.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 2531, *Ductile iron pipes, fittings, accessories and their joints for water applications*

ISO 4016, *Hexagon head bolts — Product grade C*

ISO 4034, *Hexagon nuts — Product grade C*

ISO 4633, *Rubber seals — Joint rings for water supply, drainage and sewerage pipelines — Specification for materials*

ISO 6506-1, *Metallic materials — Brinell hardness test — Part 1: Test method*

1) 100 kPa = 1 bar = 0,1 MPa; 1 MPa = 1 N/mm^2.

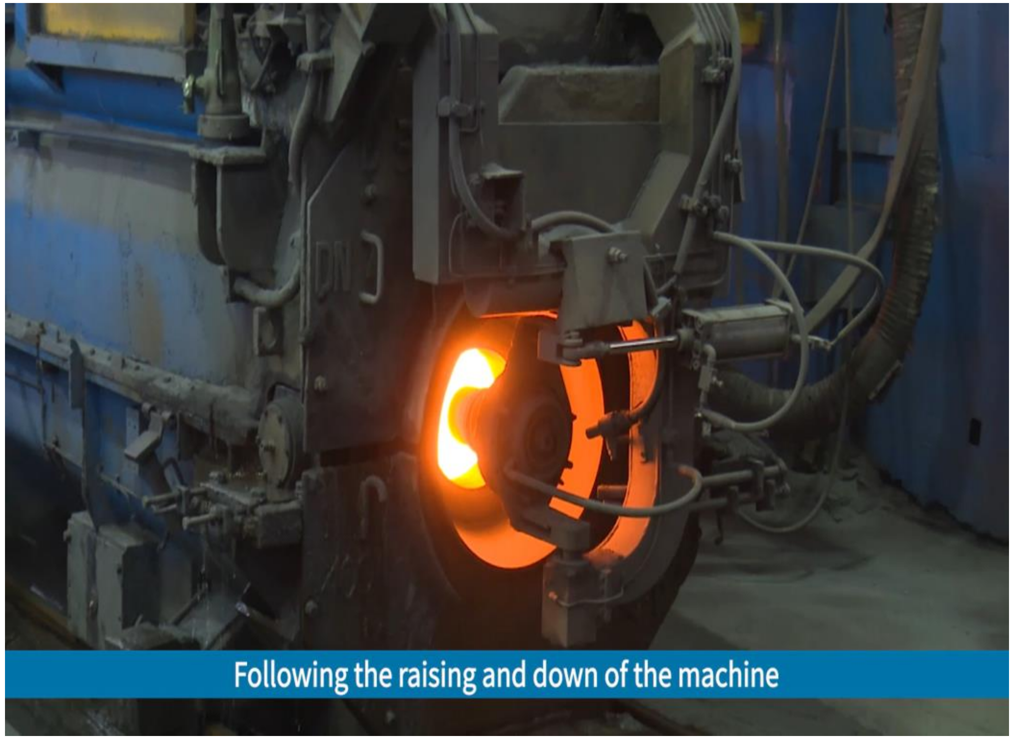
 1

PAM： FOUNDED IN 1856, A BRANCH OF SG GROUP

PAM: The Tech leading company in the world for DIP

- Auto Centrifugation Casting (year 1927)
  - ☞ Drop the cost significantly
  - ☞ Production output from 1psc/hour to 40psc /hour (DN600 as example)

→ Good pipe can be widely used for ordinary people



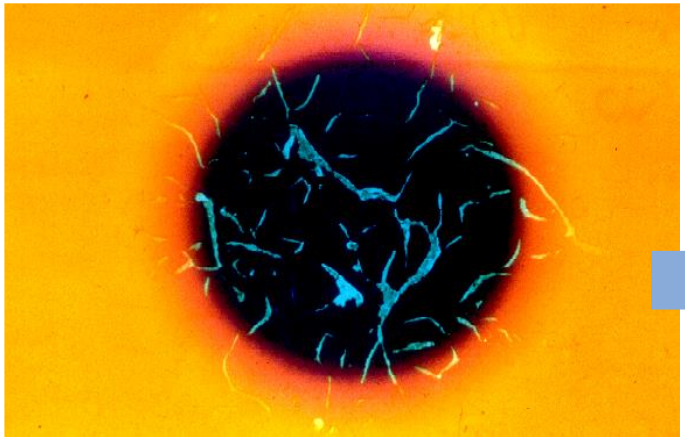
PAM SINCE 1856 穆松桥

PAM: FOUNDED IN 1856, A BRANCH OF SG GROUP

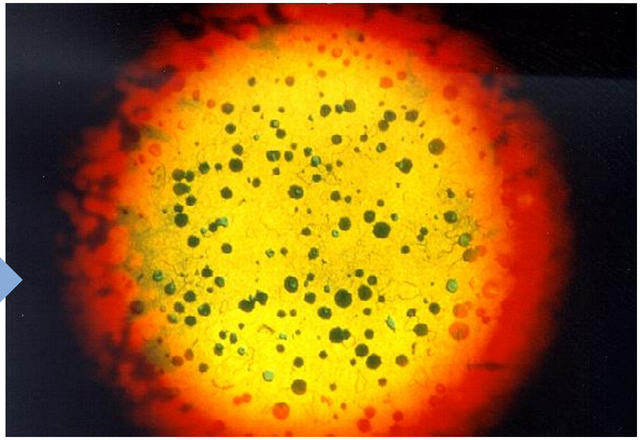
PAM: The Tech leading company in the world

- Graphite nodularization (year 1950)
  - Still iron, but have better mechanical properties than Q235 steel
  - Cost much lower than steel thanks to get rid of further metallurgy for transforming iron to steel

→ Provide better properties than steel but with much lower cost



Grey cast iron



Nodular cast iron

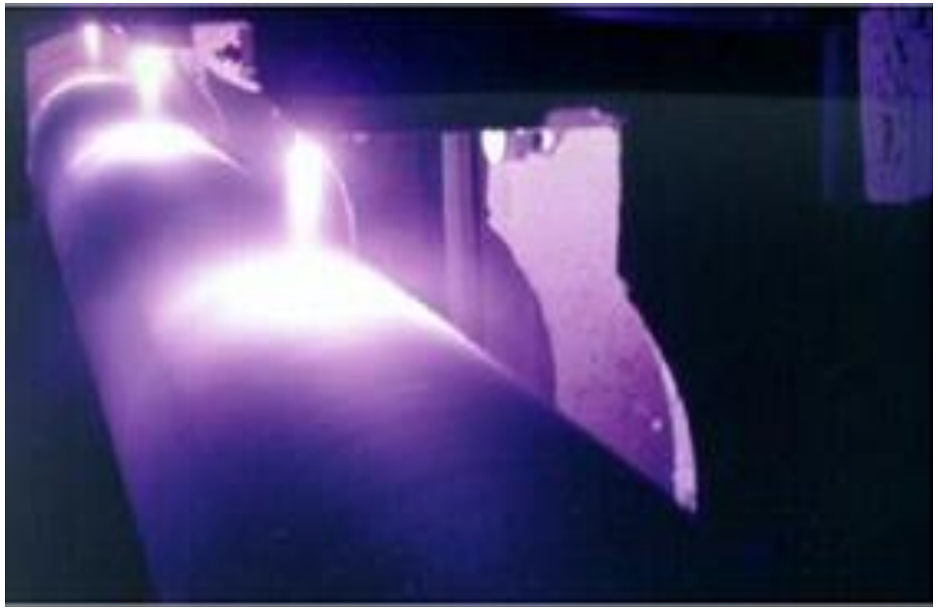
	球墨铸铁	普通钢
抗拉强度	420 ~ 500 Mpa	360 Mpa
屈服强度	270 ~ 300 Mpa	235 Mpa
延伸率	12 ~ 20 %	22 ~ 28 %
径向刚度	半刚性	柔性
疲劳强度	好	
温度敏感性	好	好
抗冲击性	好 (300~ 400 Mpa)	好

PAM： FOUNDED IN 1856, A BRANCH OF SG GROUP

PAM: The Tech leading company in the world

- Arc Zinc spraying on the pipe (year 1970)
  - ☞ Greatly improve the production efficiency vs other technology
  - ☞ Provide the excellent anti-corrosion performance

  → At least more than 50 years life time



These 3 technologies build foundation of modern ductile iron pipe

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The key milestone of PAM history

1856年	1866年	1870年	1927年	1931年	1933年	1950年	1970年	1972年
穆松桥公司成立	生产第一根铸铁管	第一批出口	获得离心技术专利许可	进口8000吨管道到南京	建立研发中心	获得铸铁球化工艺专利许可	实现铸铁球化工艺工业化	生产大口径管道

PAM
SAINT-GOBAIN

1980年	1997年	2001年	2002年	2005年	2008年	2016年	2020年
管道业务部全球扩展	进入中国 成立马鞍山基地	PAM 品牌全球启用 PAM	加速特种管道的研发	收购徐钢 成立徐州基地	整合可持续发展计划	MAS基地定制出口科威特管	日照合作生产基地

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The leading technologies

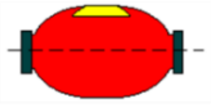
- Ultrasonic auto thickness measurement machine；
- Auto casting system；
- Trenchless ZMU coating；
- ZnAl anti-corrosion coating；
- Blast furnace slag cement for cement lining；
- Auto leakage detection system；
- Zero VOC water based paint；
- Digital twin system for production

> 1500 patents

杨松桥

PAM: FOUNDED IN 1856, A BRANCH OF SG GROUP

PAM SINCE 1856 穆松桥



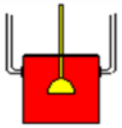
1) 中频炉

Liquid iron

1）Raw materials control

2）monitor chem compo

3）measure temp of liquid iron

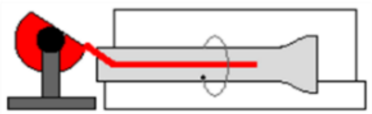


2) 球化处理

Mg treatment= nodularization

1）monitor chem compo of ladle；

2）measure the temp of liquid iron

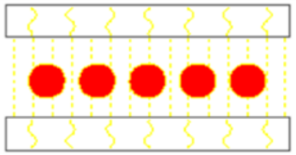


3) 离心机

Centrifugation casting

1）The highest productivity in the word；

2）**Auto casting system**

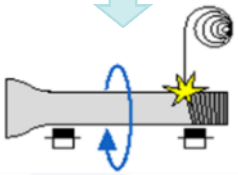


4) 热处理

Heat treatment

1）smart temp monitoring

2）Microstructure and mechanical

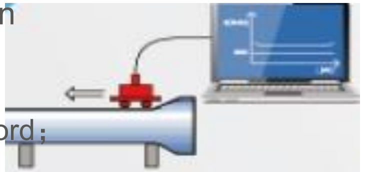


5) 喷锌

Zinc coating

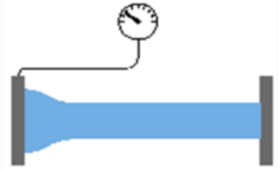
1）control zinc loading

2）**ZnAl coating**



Online wall thickness measurement

100% testing

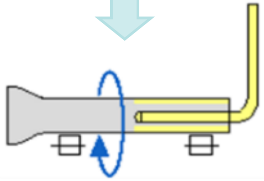


6) 水压实验

Water pressure test

1）100% test

2）**Auto leakage detection system on small pipes**

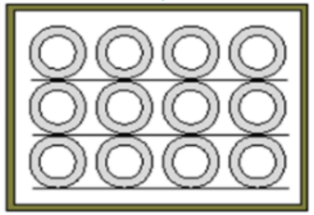


7) 水泥内涂

Cement lining

1）**Blast furnace cement**

2）auto monitor the quality

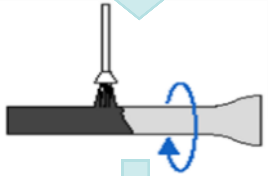


8) 养生

Cement lining curing

1）High temp steam

2）smart temp auto control



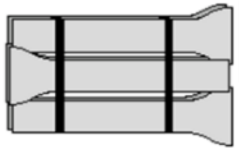
9) 沥青喷涂

External coating

1）preheating and drying stoves

2）smart temp control

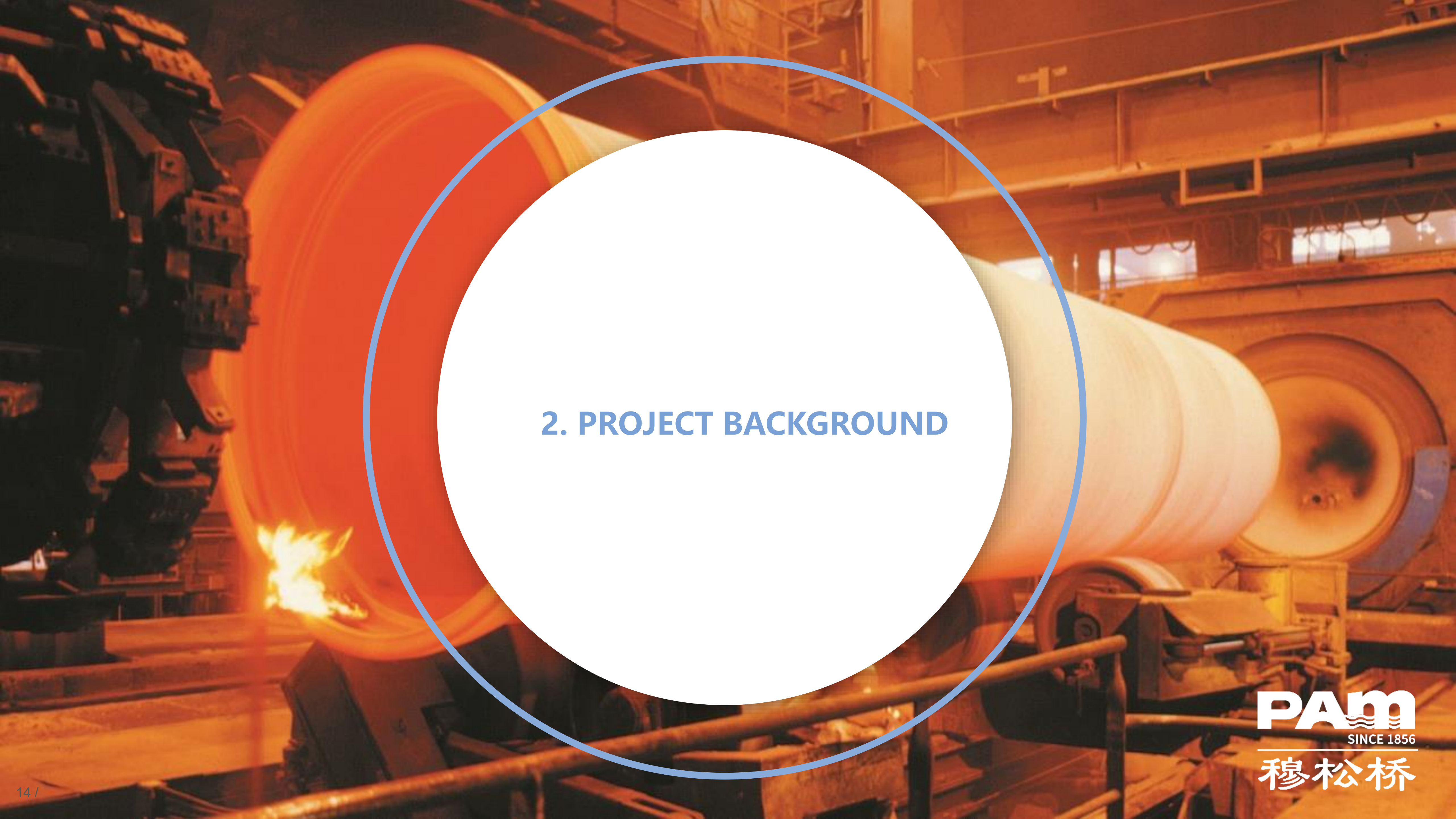
3）**VOC free water based paint**



10) 成品库

Yard

1）WMS to manage

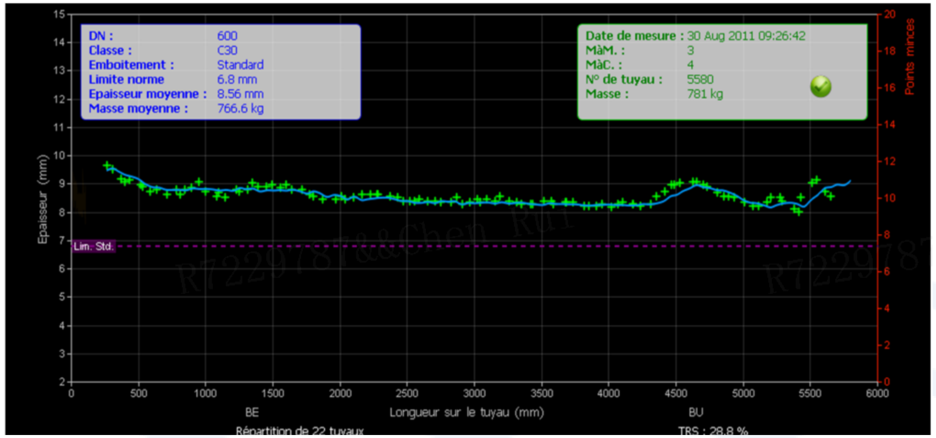


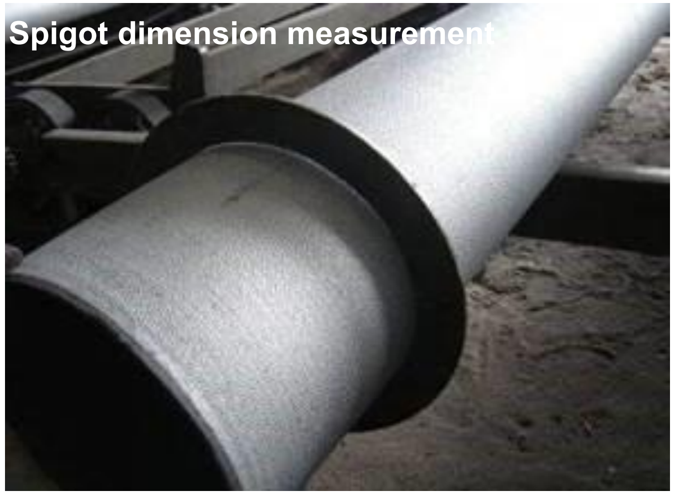
2. PROJECT BACKGROUND

PAM SINCE 1856 穆松桥

Main 3 quality controls defined in the standard

- Mechanical strength: tensile strength / elongation
- Dimension control: Socket and spigot dimension
- Wall thickness control





PROJECT BACKGROUND

Wall thickness: pipe weight and thin rejection

- Automatic ultrasonic measurement machine measures each pipe wall thickness
- Thin rejection judged if thickness below min thickness defined in standard
- Each pipe weighed and linked with thickness measurement results / rejection

Important cost kept in mind

- Liquid iron = 70% of pipe cost = 3040cny/t
  → save as much as possible but to meet min thickness requirement defined in the standard
- Thin rejection pipes to be recycled = 1040 cny/t transformation cost loss

It is an experience:
With the reduction of pipe weight, the risk of thin rejection rate increases

This project is to (under current process conditions)

- Statistically study relation curve of pipe weight vs thin rejection
- Explore if feasible to find the optimized point of pipe weight to maximize the gain
- Design software blueprint to develop software to make statistic analysis easily and rapidly

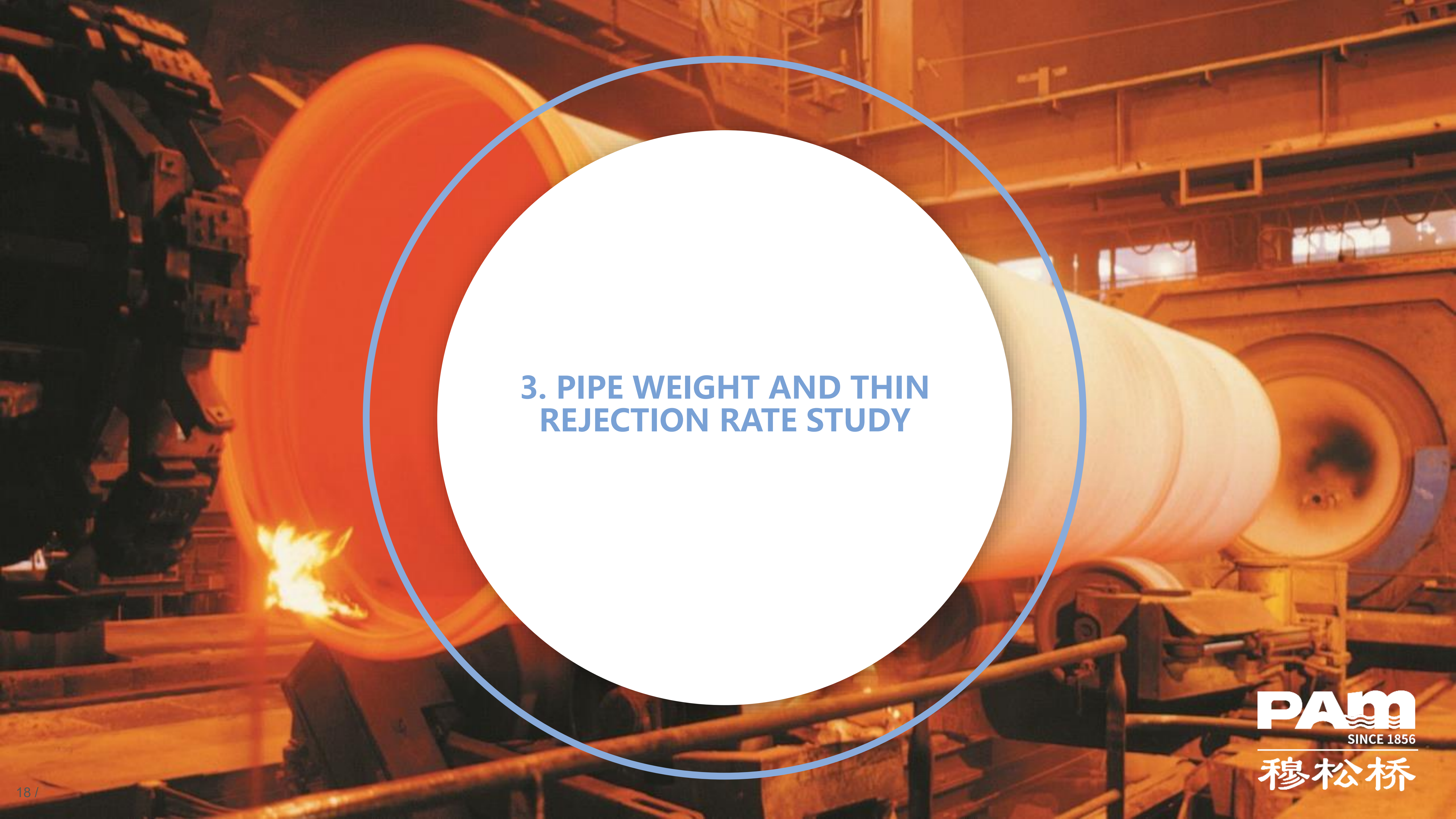
PROJECT BACKGROUND

Internship company

- Saint-Gobain Pipelines Co., Ltd
- Located in Ma'Anshan city, Anhui Province, China
- Belongs to PAM France

Group member

- CHEN Bohan: internship student from UC Davis
  - ☞ Collect data and use excel to do analysis
  - ☞ Calculate to see if feasible to find cost optimized pipe weight
  - ☞ Design the software blueprint
  - ☞ Communicate with SG Shanghai R&D center for software development by using blueprint
- WANG Jifen: Senior R&D engineer in Saint-Gobain Pipelines Co., ltd
  - ☞ As a supervisor to guide / support what CHEN Bohan is doing
- WANG Zhiyi: Software development Engineer in Saint-Gobain Shanghai Research Center (SGRS)
  - ☞ Develop the software according software blueprint
  - ☞ Frequent communication during members is necessary



3. PIPE WEIGHT AND THIN REJECTION RATE STUDY

PAM
SINCE 1856
穆松桥

PIPE WEIGHT AND THIN REJECTION RATE STUDY

Term and definition

- Pipe nominal thickness
  - Pipe nominal thickness and min thickness defined in the standard
  - Pipe nominal thickness to calculate nominal weight no matter the real pipe weight is
  - Nominal weight just to be used for all the pipe manufacturers and clients to money settlement

- Pipe nominal weight
  - Nominal weight = nominal thickness to calculate the volume * density of iron

- Iron saving
  - (Pipe nominal weight- pipe real weight) / pipe nominal weight

- Thin rejection rate
  - Number of thin pipe rejection / number of total pipe measured

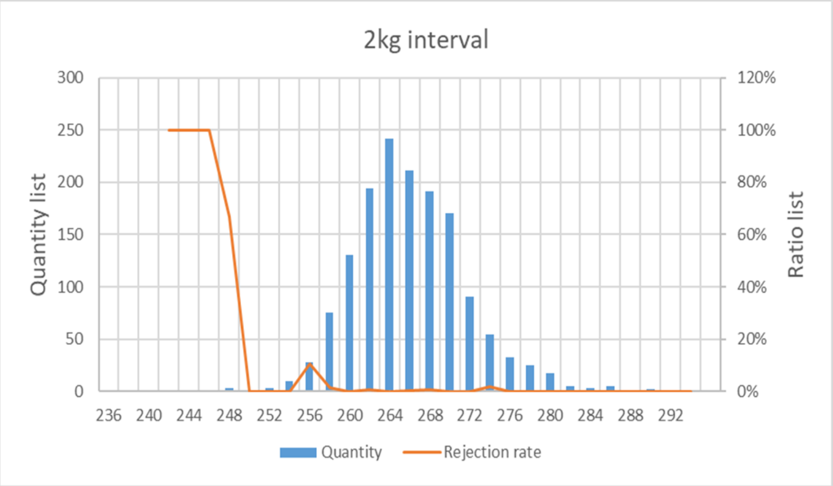
- Gain
  - (Total pipe real weight – total pipe number* pipe nominal weight) * (liquid iron cost cny/t)– total pipe rejection weight *(transformation cost cny/t)

PIPE WEIGHT AND THIN REJECTION RATE STUDY

穆松桥

1500 sample size study

- Sample information
  - ☞ Product reference: DN300, K9, TYT
  - ☞ Production date : 5/5/2023- 5/8/2023
  - ☞ Data source: Ultrasonic Thickness Gauge Backend Database

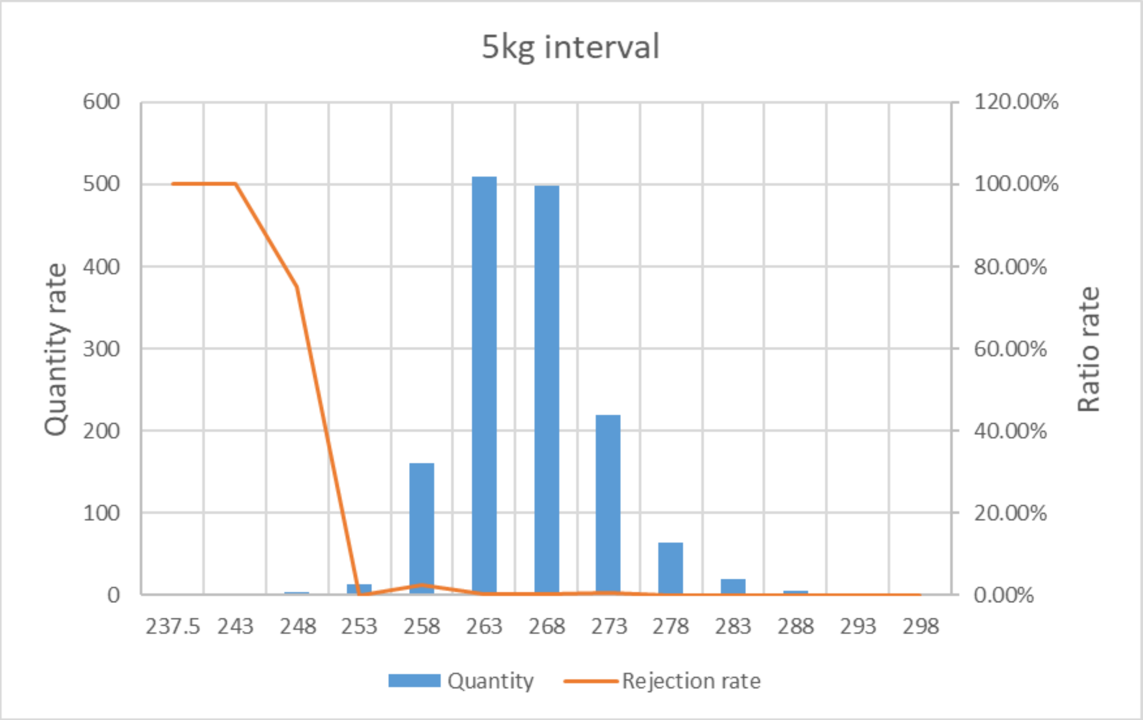


- Study method
  - ☞ Export target sample data from existing data source
  - ☞ Copy target sample data to excel and make excel table
  - ☞ Use excel related analysis tooling and calculation formula to do analysis and make chart
  - ☞ Different kg interval as group studied
    1. Count the total pipe qty (psc) in each pipe weight interval, count total rejection qty (psc) in each pipe weight interval
    2. X-axis the pipe weight with weight interval; Y-axis is total pipe qty and rejection rate in each pipe weight interval

PIPE WEIGHT AND THIN REJECTION RATE STUDY

1500 sample size study

- Results



The chart left shows:

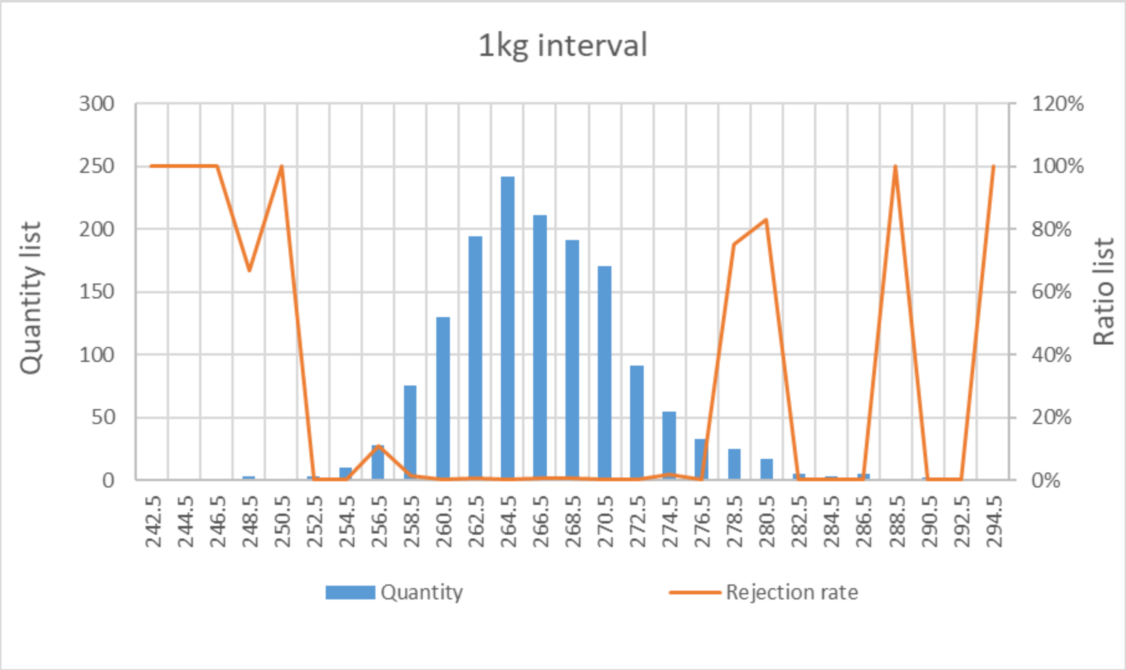
- Gaussian distribution really not obvious
- Pipe weight more than 253kg, Thin rejection rate quite low

→ Next step to reduce 5kg interval to 1kg interval

PIPE WEIGHT AND THIN REJECTION RATE STUDY

1500 sample size study

- Results



The chart left shows:

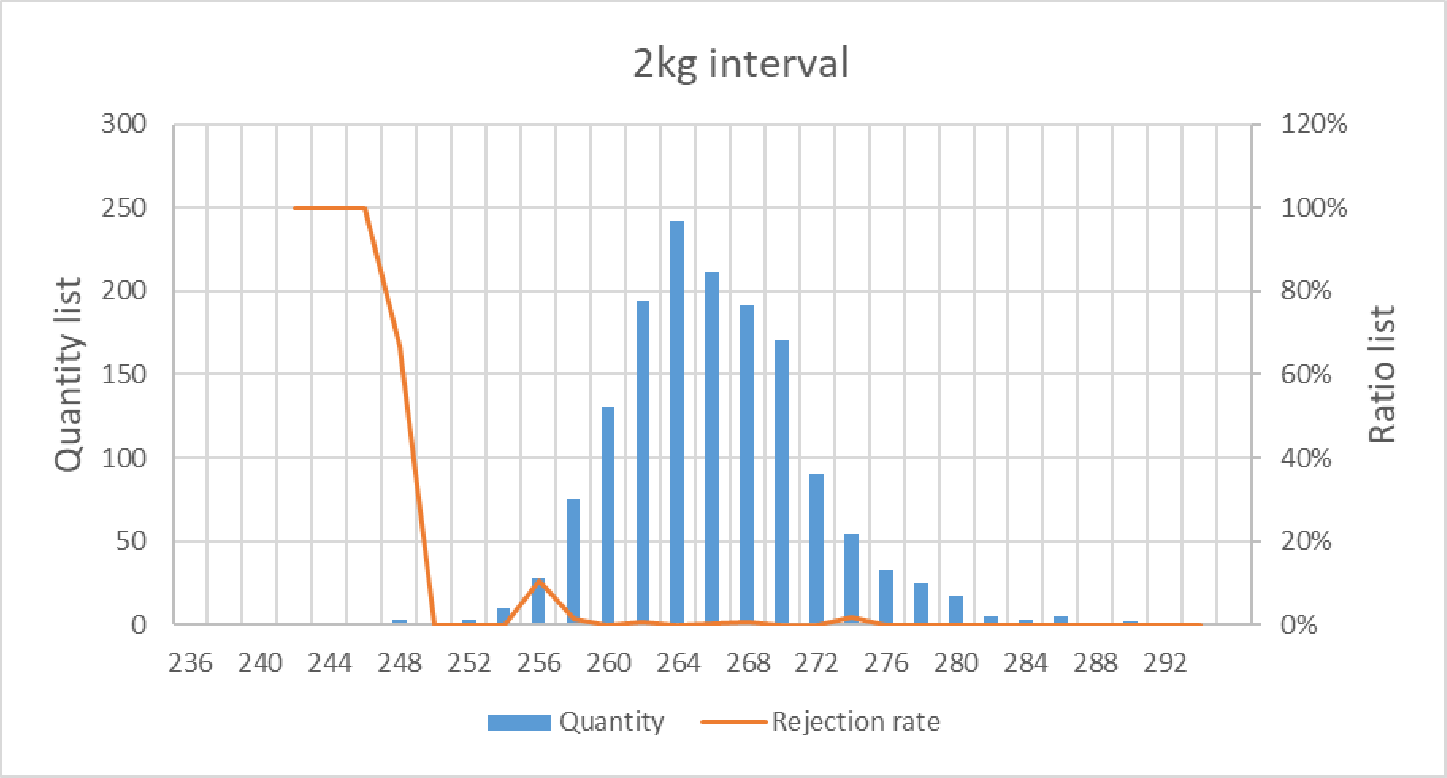
- Gaussian distribution much more obvious than that of 5kg interval, but still not perfect
- The thin rejection rate varies a lot on the both ends of chart, varying even from 0% to 100%

→ Try 2kg interval to see the changes (between 5kg and 1kg)

PIPE WEIGHT AND THIN REJECTION RATE STUDY

1500 sample size study

- Results



The chart left shows:

- The gaussian distribution still not perfect, but much better than 5kg interval
- Thin rejection rate becomes more regular than 1kg interval

PIPE WEIGHT AND THIN REJECTION RATE STUDY

Analysis and discussion

- All 3 pipe weight intervals show that Gaussian distribution not perfect, possible reasons
  - ☞ Man: new nonskilled workers / some data deleted from the data / some data not recording
  - ☞ Machine: casting machine and weight machine not on the right condition
  - ☞ Material: liquid iron viscosity changed
  - ☞ Method: Target weight changed
  - ☞ Environment
  - ☞ Sample size is not enough

  → After investigation, there is no change for man, machine, material, method and environment, so the sample size enlargement to be tried

- Different pipe weight intervals have effect on the thin rejection rate greatly, the reasons
  - ☞ If interval is too big, Gaussian distribution is not obvious / the definition of thin rejection is to good
  - ☞ If interval is too small, Gaussian distribution becomes better, if one pipe in this interval and also this pipe is thin rejected, the thin rejection rate 100%

  → Choose the right pipe interval very important / should be set on the UI to let users input

PIPE WEIGHT AND THIN REJECTION RATE STUDY

PAM SINCE 1856 穆松桥

10000 sample size study

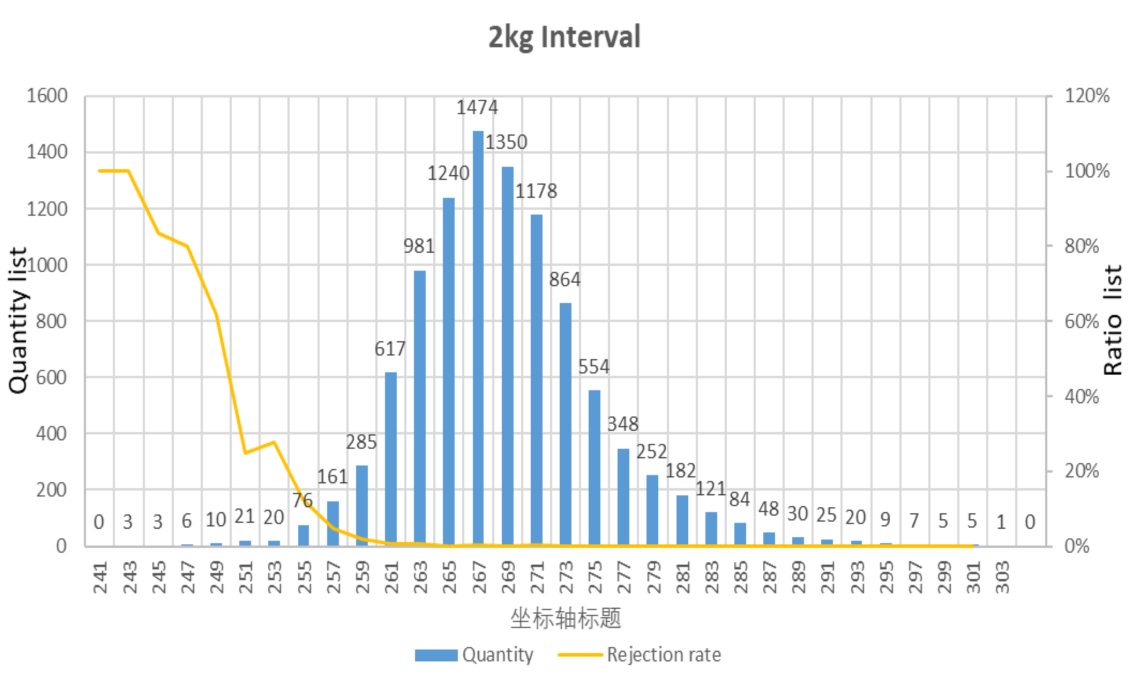
- Sample information
  - Product reference: DN300, K9, TYT
  - Production date: 3/27/2023- 5/15/2023
  - Data source : Ultrasonic Thickness Gauge
  - Backend Database

- Study tooling
  - Export target sample data from existing data source
  - Copy target sample data to excel and make excel table
  - Use excel related analysis tooling and calculation formula to do analysis and make chart
  - Different kg interval as group studied
    1. Count the total pipe qty (psc) in each pipe weight interval, count total rejection qty (psc) in each pipe weight interval
    2. X-axis the pipe weight with weight interval; Y-axis is total pipe qty and rejection rate in each pipe weight interval

PIPE WEIGHT AND THIN REJECTION RATE STUDY

10000 sample size study

- Results



Weight	>	≤	Reject rate	Saving iron rate
241	240	242	100.00%	0%
243	242	244	100.00%	0%
245	244	246	100.00%	0%
247	246	248	83.33%	23.22%
249	248	250	80.00%	22.60%
251	250	252	61.90%	22.14%
253	252	254	25.00%	21.44%
255	254	256	27.63%	20.83%
257	256	258	12.42%	20.23%
259	258	260	4.91%	19.60%
261	260	262	1.78%	19.00%
263	262	264	0.71%	18.39%
265	264	266	0.81%	17.78%
267	266	268	0.14%	17.15%
269	268	270	0.22%	16.53%
271	270	272	0.08%	15.93%
273	272	274	0.35%	15.30%
275	274	276	0.00%	14.69%
277	276	278	0.00%	14.08%
279	278	280	0.00%	13.45%

The left chart shows :

- The distribution is nearly the complete Gaussian distribution
- The pipe weight more than 256kg, thin rejection rate quite low.
- The thin rejection rate curve becomes more smooth.

Analysis and discussion

- The 10000 sample size 2kg interval chart show that the Gaussian distribution is nearly perfect
- The Gaussian distribution cannot be 100 percent perfect, it can just be nearly perfect.
- It would be better to have bigger sample size to make the analysis more reliable if possible.

PIPE WEIGHT AND THIN REJECTION RATE STUDY

Gain analysis table- how to calculate

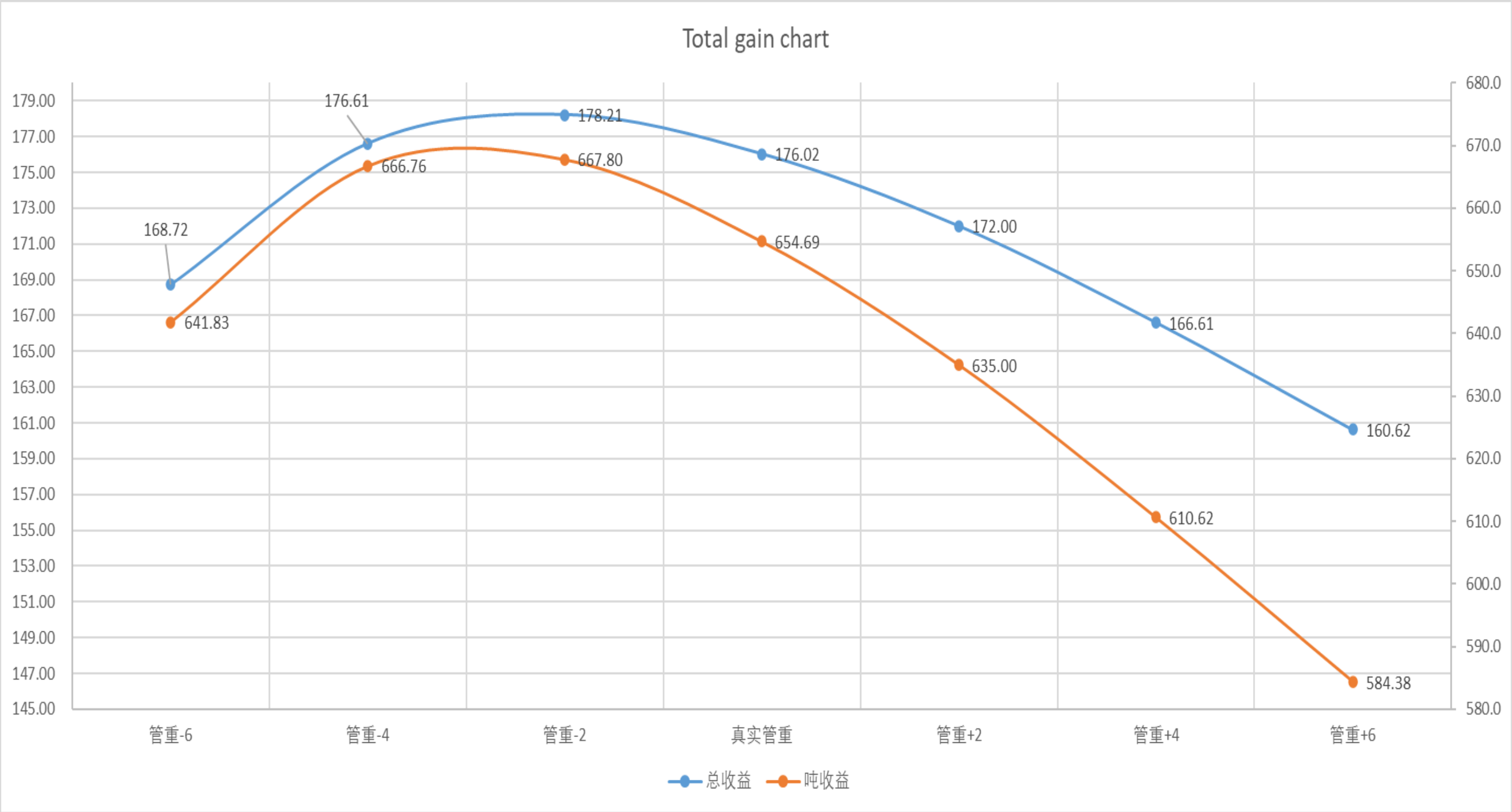
Weight	>	≤	Number	废品率	节铁吨数	废品吨数	3400	1020	成本差
243	242	244	3	**100.00%**	**0.00**	**0.49**	0.00	0.05	-0.05
245	244	246	3	**100.00%**	**0.00**	**0.49**	0.00	0.05	-0.05
247	246	248	6	**83.33%**	**0.08**	**1.03**	0.03	0.11	-0.08
249	248	250	10	**80.00%**	**0.15**	**1.80**	0.05	0.18	-0.13
251	250	252	21	**61.90%**	**0.57**	**3.12**	0.19	0.32	-0.12
253	252	254	20	**25.00%**	**1.04**	**1.21**	0.35	0.12	0.23
255	254	256	76	**27.63%**	**3.70**	**5.30**	1.26	0.54	0.72
257	256	258	161	**12.42%**	**9.21**	**5.12**	3.13	0.52	2.61
259	258	260	285	**4.91%**	**17.16**	**3.62**	5.83	0.37	5.46
261	260	262	617	**1.78%**	**37.19**	**2.87**	12.64	0.29	12.35
263	262	264	981	**0.71%**	**57.86**	**1.84**	19.67	0.19	19.48
265	264	266	1240	**0.81%**	**70.64**	**2.65**	24.02	0.27	23.75
267	266	268	1474	**0.14%**	**81.54**	**0.53**	27.72	0.05	27.67
269	268	270	1350	**0.22%**	**71.92**	**0.81**	24.45	0.08	24.37
271	270	272	1178	**0.08%**	**60.56**	**0.27**	20.59	0.03	20.56
273	272	274	864	**0.35%**	**42.55**	**0.82**	14.47	0.08	14.38
275	274	276	554	**0.00%**	**26.29**	**0.00**	8.94	0.00	8.94
277	276	278	348	**0.00%**	**15.83**	**0.00**	5.38	0.00	5.38
279	278	280	252	**0.00%**	**10.95**	**0.00**	3.72	0.00	3.72
281	280	282	182	**0.00%**	**7.54**	**0.00**	2.56	0.00	2.56
283	282	284	121	**0.00%**	**4.76**	**0.00**	1.62	0.00	1.62
285	284	286	84	**0.00%**	**3.15**	**0.00**	1.07	0.00	1.07
287	286	288	48	**0.00%**	**1.70**	**0.00**	0.58	0.00	0.58
289	288	290	30	**0.00%**	**1.00**	**0.00**	0.34	0.00	0.34
291	290	292	25	**0.00%**	**0.78**	**0.00**	0.27	0.00	0.27
293	292	294	20	**0.00%**	**0.59**	**0.00**	0.20	0.00	0.20
295	294	296	9	**0.00%**	**0.25**	**0.00**	0.08	0.00	0.08
297	296	298	7	**0.00%**	**0.18**	**0.00**	0.06	0.00	0.06
299	298	300	5	**0.00%**	**0.12**	**0.00**	0.04	0.00	0.04
301	300	302	5						**176.02**

Comments:

- 3400 cny/t is the cost of liquid iron
- 1020 cny/t is the transformation cost loss for rejection pipe
- Iron saving rate= (Pipe nominal weight- pipe real weight)/ pipe nominal weight
- Thin rejection rate=Number of thin pipe rejection / number of total pipe measured

PIPE WEIGHT AND THIN REJECTION RATE STUDY

Gain simulation with pipe wight reduction



The left chart shows :

- When decrease 2kg of real pipe weight we can get the maximum total gain and per-ton revenue
- The simulation of decrease/ increase weight on pipe weight can only pan the image left and right, the thin rejection rate will not change as the pipe weight decrease or increase.

	pipe weight-6	pipe weight-4	pipe weight-2	real pipe weight	pipe weight+2	pipe weight+4	pipe weight+6
average value	263.3	265.3	267.3	269.3	271.3	273.3	275.3
standard deviation	6.53	6.53	6.53	6.53	6.53	6.53	6.53

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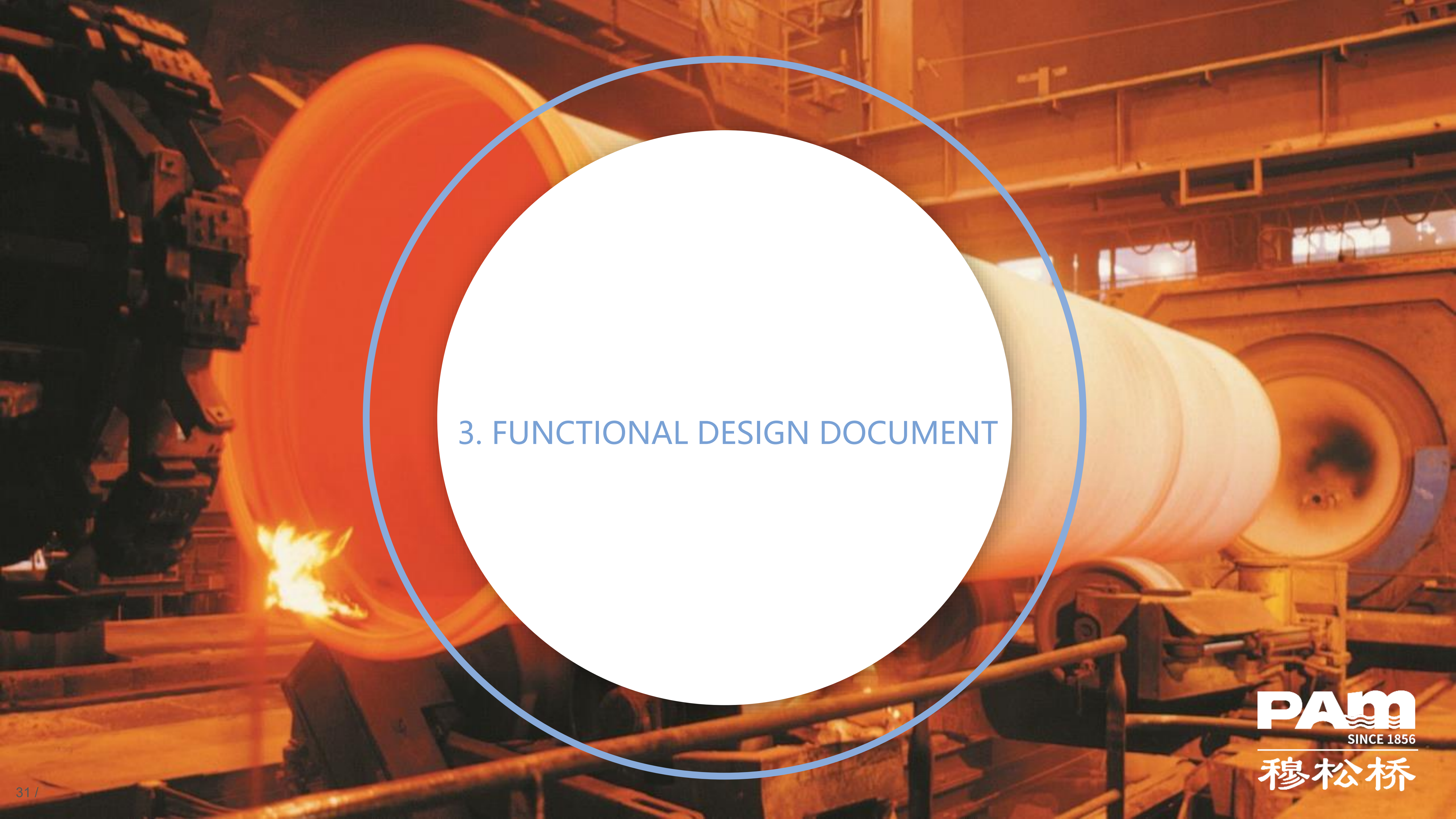
Conclusion

- By conducting a study on the relationship between pipe weight and thin rejection rate, it was shown that by finding the equilibrium between thin rejection rate and pipe weight, it is possible to reduce the company’s costs and maximize the gain.

- In the Gaussian distribution, the sample size is really important, the bigger the sample size is, the more nearly perfect the Gaussian distribution is.

- Choosing the right interval is also important to plot the Gaussian distribution, otherwise it will be difficult to find any useful details in the Gaussian distribution.



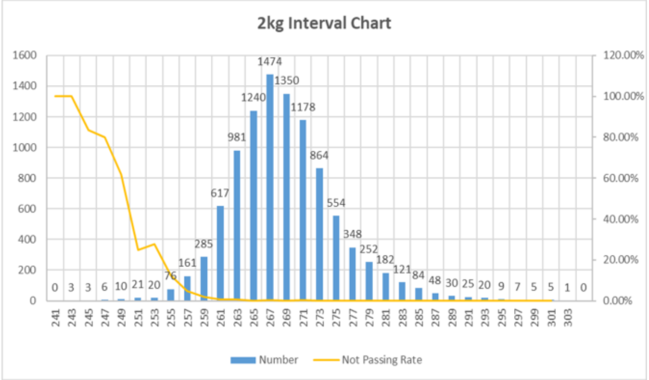
3. FUNCTIONAL DESIGN DOCUMENT

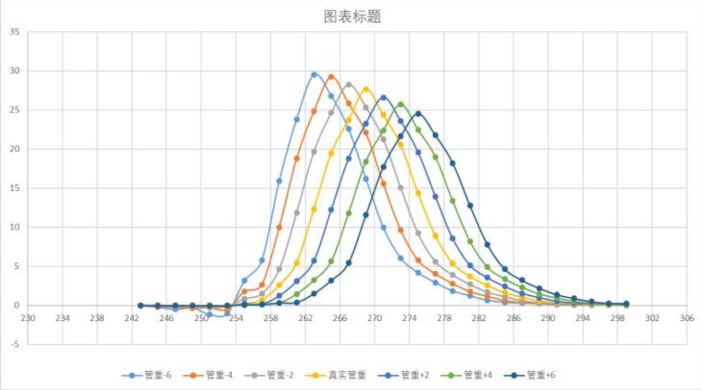
Input logic

- Input
  - Automation
    - Thickness database (pipe weights and corresponding thickness results)
    - Standard pipe weight chart
    - Number of simulated pipe weight increases and decreases (specified as three)
  - Manually
    - Starting pipe weight, ending pipe weight and pipe weight interval
    - Manually enter pipe specifications such as caliber, wall thickness, grade.
    - Difference durations: long-term(default 6 months), medium-term(default 3 months) and near-term(default 1 month)
    - Iron saving benefits, rejection consumption

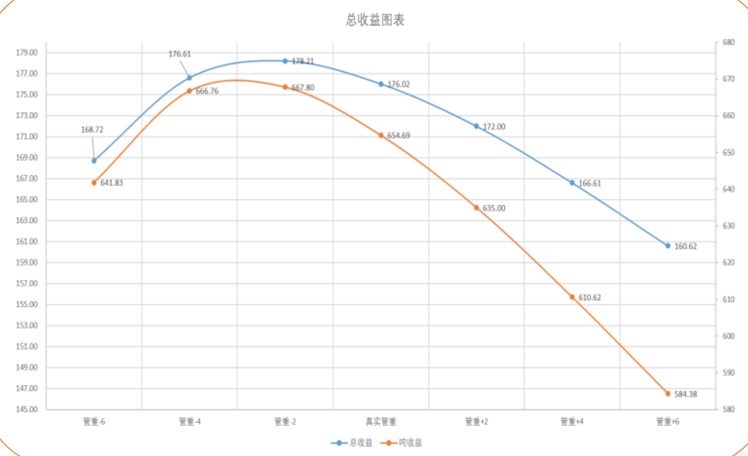
Output logic

Output





1.Graph of probability density distribution of pipe weight
2.Average pipe weight ± σ (standard deviation) chart



Charting the difference in rejection rate and iron saving rate between the real pipe weight and the simulated maximum pipe weight of the total gain

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Calculation

Calculation logic

Filter the data within a pipe weight interval as required in the thickness measurement database

Calculate the number of good and reject pipes in this pipe weight interval, number of good pipes + number of reject pipes = total number of pipes

Number of reject pipes/ Total pipes =rejection rate in the interval. Individual iron saving weight = Standard pipe weight - Individual good pipe weight , Total iron saving weight = SUM(sum of iron saving weights of all individual good pipes in the interval), Total iron saving weight in the interval = Total iron saving weight / (standard pipe weight × total number of good pipe)

Calculate the rejection rate and iron saving rate in all intervals

Average pipe weight= total pipe weight / total number of pipes
standard deviation=Sqrt((Sum (individual pipe weight - average pipe)^2)/total pipe number

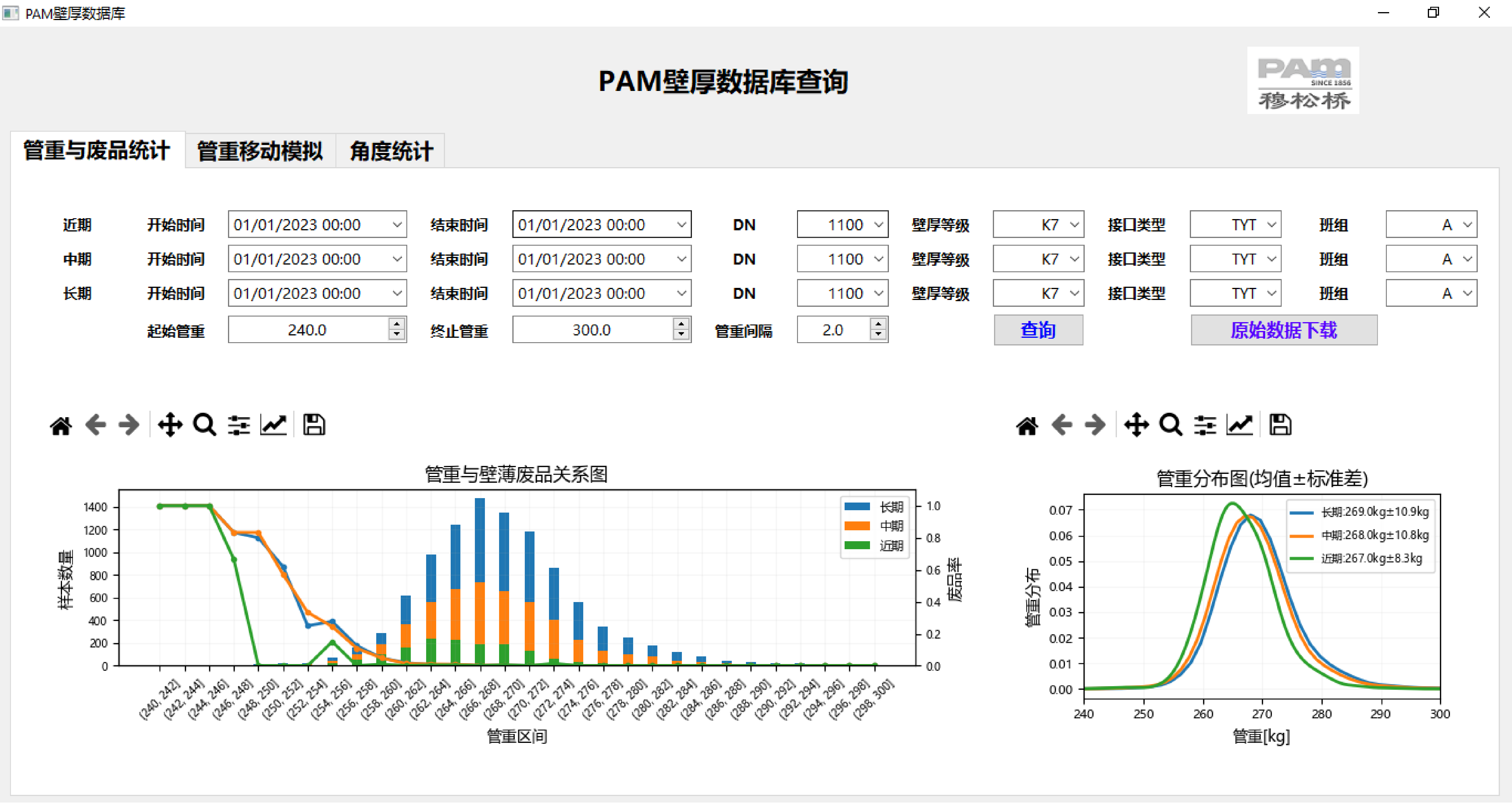
rejection rate after simulation= total weight of reject iron/ total weight of pipe
iron saving rate after simulation= total weight of saving iron/[(1-reject iron rate)*total pipe number*standard pipe weight

Filter the data within a pipe weight interval as required in the thickness measurement database

Calculates the total number of pipes in this pipe weight interval. The median value of the pipe weight within the pipe interval. All simulations are based on the rejection rate and iron saving rate of the true values.

Total weight of iron saving = Iron saving rate in the corresponding interval× standard pipe weight ×total number of pipes in the corresponding interval × good pipe rate
Total weight of reject pipe = Total pipe weight within the pipe interval× rejection rate in the corresponding interval

Cost difference = iron saving revenue ×tons of rion saved - rejection consumption cost ×tons of iron rejection


PAM壁厚数据库
PAM壁厚数据库查询
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管重与废品统计
管重移动模拟
角度统计
近期
中期
长期
开始时间
01/01/2023 00:00
结束时间
01/01/2023 00:00
DN
1100
壁厚等级
K7
接口类型
TYT
班组
A
起始管重
240.0
终止管重
300.0
管重间隔
2.0
查询
原始数据下载
管重与壁薄废品关系图
样本数量
废品率
管重区间
长期
中期
近期
管重分布图(均值±标准差)
管重分布
管重[kg]
长期:269.0kg±10.9kg
中期:268.0kg±10.8kg
近期:267.0kg±8.3kg

PAM壁厚数据库

PAM壁厚数据库查询

管重与废品统计 | 管重移动模拟 | 角度统计

输入

铁水成本 [CNY/T] 3400

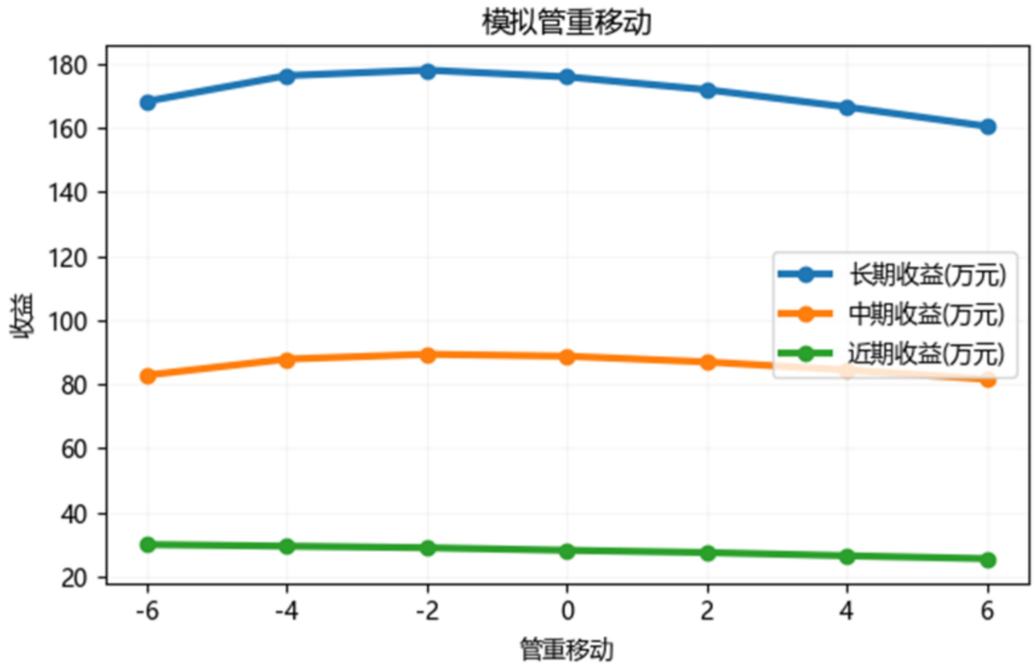
废品成本 [CNY/T] 1020

开始模拟

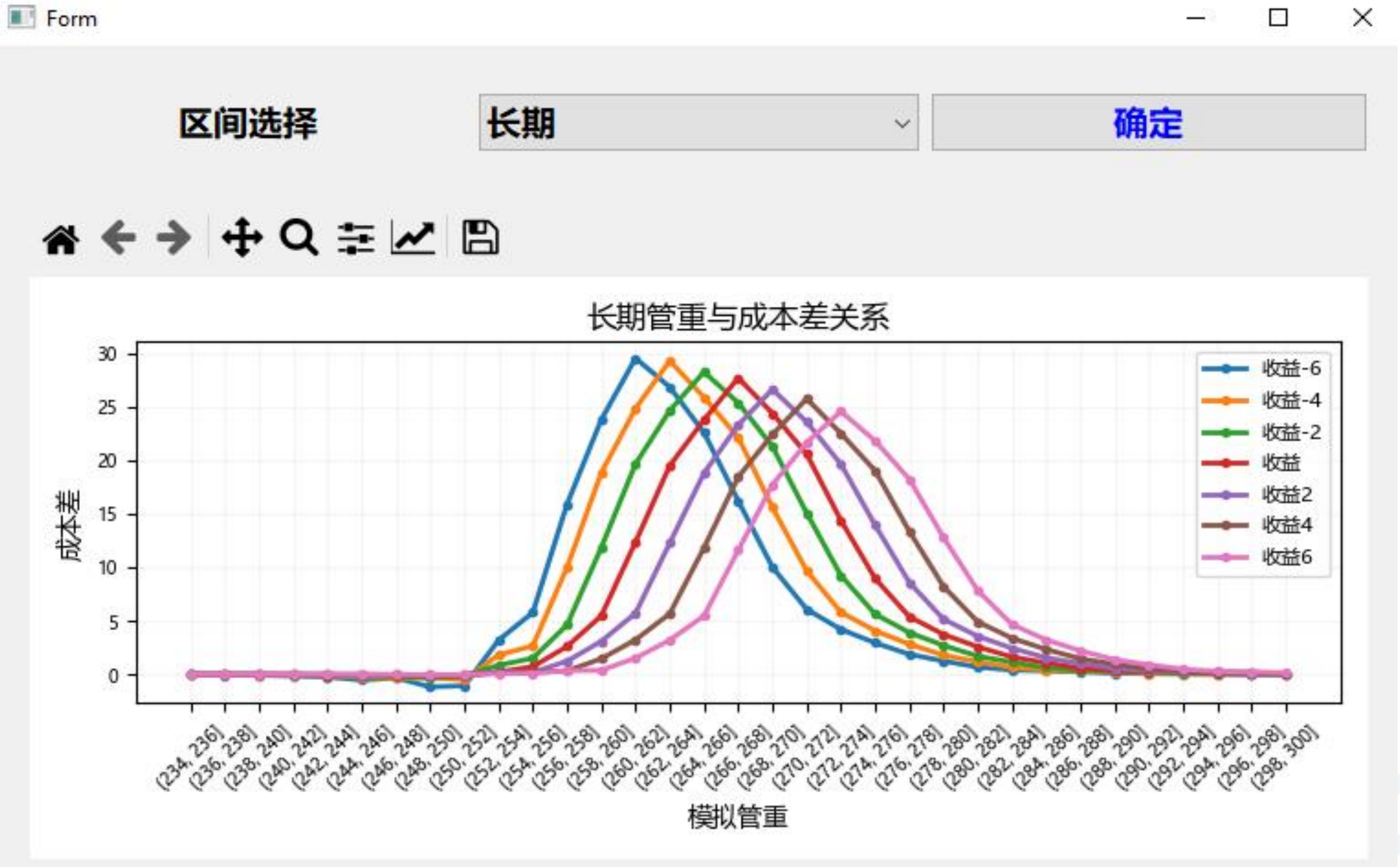
输出

管道规格	DN	1100	K7	TNT

长期	管重区间		收益		废品率		节铁率	
B=目前最佳	(266, 268]	kg	655.0	元/吨	1.2	%	16.3	%
A=模拟最佳	(264, 266]	kg	667.4	元/吨	2.2	%	16.7	%
A-B	-2	kg	12.4	元/吨	1.0	%	0.4	%
中期	管重区间		收益		废品率		节铁率	
B=目前最佳	(266, 268]	kg	662.2	元/吨	1.6	%	16.5	%
A=模拟最佳	(264, 266]	kg	671.6	元/吨	2.7	%	16.9	%
A-B	-2	kg	9.4	元/吨	1.1	%	0.4	%
近期	管重区间		收益		废品率		节铁率	
B=目前最佳	(262, 264]	kg	704.8	元/吨	0.8	%	17.3	%
A=模拟最佳	(256, 258]	kg	767.3	元/吨	2.4	%	18.8	%
A-B	-6	kg	62.5	元/吨	1.6	%	1.5	%



更多


Form
区间选择
长期
确定
长期管重与成本差关系
成本差
30
25
20
15
10
5
0
(234, 236] (236, 238] (238, 240] (240, 242] (242, 244] (244, 246] (246, 248] (248, 250] (250, 252] (252, 254] (254, 256] (256, 258] (258, 260] (260, 262] (262, 264] (264, 266] (266, 268] (268, 270] (270, 272] (272, 274] (274, 276] (276, 278] (278, 280] (280, 282] (282, 284] (284, 286] (286, 288] (288, 290] (290, 292] (292, 294] (294, 296] (296, 298] (298, 300]
模拟管重
收益-6
收益-4
收益-2
收益
收益2
收益4
收益6

FINAL SUMMARY AND REFLECTIONS

Summary and reflection

- Learned how to use Excel to create tables and have ability to use most common formulas
- Learned how to make a standardized and beautiful PowerPoint
- Learned the meaning of average, standard deviation of statistic and the statistic real application
- Learned how to use statistic and math to analyze the data
- Knowing that the database is the pre-condition of statistic analysis and database must be 1st collected and created in an appropriate way
- Knowing that the database is useless if don’t use it and let them sleep there and it is important to use the database to make analysis to help the business
- Knowing that it is difficult to use excel table to do analysis and complicated calculation and it is important to use software to program to relieve the workload

THANKS FOR YOUR ATTENTION!

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